MPECSS: Scholtes Regularization with Adaptive Paths for Targeting B-stationary Points in MPECs

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ABSTRACT

We propose MPECSS (MPEC Stationarity Search), an open-source algorithm for solving Mathematical Programs with Equilibrium Constraints (MPECs), which arise in bilevel optimization, optimal control, and equilibrium modeling. Standard Scholtes regularization converges only to C-stationary points in general, a comparatively weak optimality concept providing no systematic mechanism for certifying the stronger B-stationary solutions required for genuine local optimality. MPECSS addresses this limitation through two core mechanisms: an adaptive regularization-parameter update driven by complementarity residuals, and a multiplier sign test that dynamically steers the solver toward B-stationary points. Implemented in Python via CasADi and IPOPT, with no commercial dependencies, the implementation is fully open-source and reproducible. Convergence analysis establishes C-stationarity under standard assumptions; furthermore, MPECSS systematically certifies B-stationarity either directly when the multiplier sign test confirms S-stationarity under MPEC-LICQ or alternatively through LPEC (Linear Program with Equilibrium Constraints) verification. Benchmarking on MacMPEC, MPECLib, and NOSBENCH achieves convergence rates of 96.3%, 92.4%, and 93%, respectively, positioning MPECSS as an open-source method evaluated across all three major MPEC benchmark collections.

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1. Introduction

Mathematical Programs with Equilibrium Constraints (MPECs), equivalently referred to as Mathematical Programs with Complementarity Constraints (MPCCs),[1] are optimization problems [32, 37, 41] that arise in a wide range of applications, including bilevel optimization [5, 8], noncooperative game theory [21, 30], contact mechanics [27, 37], traffic flow equilibrium [13, 33], nonsmooth optimal control [36], and chemical engineering process optimization [4].

Luo et al. [32] established the formal study of MPECs and formulated their foundational theory. In general, a standard MPEC can be formulated as follows [45]:

$$\begin{aligned} \min_{x\in\mathbb{R}^n} \quad & f(x) \\ \text{subject to} \quad & g_i(x) \le 0, \quad i = 1,\dots,m, \\ & h_j(x) = 0, \quad j = 1,\dots,p, \\ & G_\ell(x) \ge 0, \quad H_\ell(x) \ge 0, \quad \ell = 1,\dots,q, \\ & G_\ell(x)^\top H_\ell(x) = 0, \quad \ell = 1,\dots,q. \end{aligned} \tag{1}$$

Here, $f(x)$ is the objective function, $g_i(x)$ and $h_j(x)$ represent the standard inequality and equality constraints respectively.

The fundamental difficulty arising from complementarity constraints is that standard constraint qualifications such as Linear Independence Constraint Qualification (LICQ) and Mangasarian–Fromovitz Constraint Qualification (MFCQ) are violated at all feasible points [41]. Consequently, classical KKT condition is failed, and standard Nonlinear Programming (NLP) solvers applied directly to (1) will not find

[1]While the literature often uses MPEC and MPCC interchangeably, we adopt the term MPEC throughout this paper for consistency.

valid solutions. To overcome this computational barrier several specialized algorithms have been developed over the past two decades. Notable developments include regularization techniques [23, 31, 42, 43], sequential quadratic programming (SQP) approaches [16, 17], penalty methods [2, 20], and interior-point strategies [28, 38].

The most widely used approach for solving MPECs is *Scholtes regularization* [42], which replaces the hard complementarity product $G_\ell(x) \cdot H_\ell(x) = 0$ with the relaxed condition $G_\ell(x) \cdot H_\ell(x) \leq t_k$, and drives $t_k \to 0$ by solving a sequence of smooth NLP subproblems. While numerically robust and widely applicable, the standard formulation reduces t_k via a fixed geometric rule that is independent of solver progress, and is theoretically guaranteed to converge only to C-stationary points [19, 42], providing no built-in mechanism for targeting B-stationary. A detailed review of Scholtes regularization is given in Section 3.1; see also Hoheisel et al. [19] for a comprehensive theoretical and numerical comparison among different relaxation methods.

Existing methods face practical limitations: SQP approaches [16] often depend on commercial solvers like SNOPT [18], while penalty [2, 20] and interior-point methods [28, 38] frequently encounter numerical instability near the feasible boundary. The recent MPECopt algorithm of Nurkanović and Leyffer [35] is an open-source method with global convergence to B-stationary points and strong performance on the MacMPEC (191 problems) benchmark [29]; however, its implementation depends on the commercial MATLAB environment and its strongest results rely on the commercial solver Gurobi, and it has not yet been validated on the MPECLib (92 problems) [9] or NOSBENCH (603 problems) [36] collections.

Significant gaps remain in the practical MPEC software landscape. Many algorithms rely on commercial software (HSL, SNOPT, Gurobi), standard Scholtes regularization lacks built-in mechanisms for diagnosing stronger stationarity than C-stationarity, and reproducible open-source benchmarking infrastructure is still comparatively scarce. MPECSS is designed to close these implementation and reproducibility gaps within a single open-source code base.

MPECSS uses the Scholtes framework while adding an implementation-aware *adaptive regularization update* driven by the homotopy complementarity residual and an integrated *multiplier sign test* derived from the mathematical characterizations of Scheel

and Scholtes [41]. In the implementation, this sign test is used as an S-stationarity diagnostic on the G- and H-multipliers at numerically biactive indices; final B-stationarity certification is then performed in Phase III by a combination of MPEC-LICQ checks and, when needed, LPEC-based verification. The method is implemented entirely in Python using the open-source CasADi framework [1] and IPOPT [44], with no commercial dependencies.

The main contributions of this paper are summarized as follows: *MPECSS algorithm:* A fully open-source, multi-phase algorithm for solving MPECs that combines Scholtes regularization with adaptive parameter control, a multiplier-based stationarity diagnostic, multistart and restoration mechanisms, and a Phase III certification pipeline combining LICQ checks, BNLP polishing, and LPEC refinement. *Adaptive regularization control:* A monitoring-driven update of t_k based on the implemented homotopy residual, stable biactive-set tracking, and stagnation counters, together with practical safeguards such as adaptive jumps, jump caps, and a floor on t_k. *Multiplier sign test:* A fast algebraic procedure that checks the converted G- and H-multipliers on numerically biactive indices to detect S-stationarity candidates and steer the subsequent certification logic. *Open-source benchmark infrastructure:* A reproducible benchmark runner for MacMPEC, MPECLib, and NOSBENCH with 249-column CSV logging, environment capture, process-level isolation for per-problem timeout enforcement, resume support (via `--resume`), retry-failed logic, and graceful recovery from timeout, interrupt, and out-of-memory events. The solver stack auto-selects between SQP+qpOASES (for problems with $n \leq 400$ when available) and IPOPT+MUMPS (otherwise). *Rigorous convergence guarantees:* We provide formal proofs that MPECSS inherits the baseline C-stationarity guarantees of Scholtes regularization, while stronger B-stationarity certificates are produced in Phase III when LICQ equivalence or LPEC verification succeeds.

2. Preliminaries

In this section we establish the notation and definitions used throughout the paper, recall the stationarity concepts and relevant constraint qualifications for MPECs.

2.1. Index sets and the MPEC Lagrangian

Throughout this section, and for the remainder of the paper, we assume that f, g_i $(i = 1, \ldots, m)$, h_j $(j = 1, \ldots, p)$, G_ℓ, and H_ℓ $(\ell = 1, \ldots, q)$ are twice continuously differentiable.

Consider the MPEC (1). We denote its feasible set by

$$\mathcal{F} = \big\{x \in \mathbb{R}^n : g_i(x) \leq 0,\ h_j(x) = 0,\ G_\ell(x) \geq 0,\ H_\ell(x) \geq 0,\ G_\ell(x)\, H_\ell(x) = 0,\ \forall\, i, j, \ell\big\}. \tag{2}$$

Let $x^* \in \mathcal{F}$ be a feasible point. Following standard MPEC literature [32, 41], we partition the active complementarity components into the following index sets at x^*:

$$\mathcal{I}_g(x^*) = \{i \in \{1, \ldots, m\} : g_i(x^*) = 0\}, \tag{3}$$

$$\mathcal{I}_{0+}(x^*) = \{\ell \in \{1, \ldots, q\} : G_\ell(x^*) = 0,\ H_\ell(x^*) > 0\}, \tag{4}$$

$$\mathcal{I}_{+0}(x^*) = \{\ell \in \{1, \ldots, q\} : G_\ell(x^*) > 0,\ H_\ell(x^*) = 0\}, \tag{5}$$

$$\mathcal{I}_{00}(x^*) = \{\ell \in \{1, \ldots, q\} : G_\ell(x^*) = 0,\ H_\ell(x^*) = 0\}. \tag{6}$$

The set $\mathcal{I}_{0+}$ contains the indices where only G is active; the set $\mathcal{I}_{+0}$ contains the indices where only H is active; and the *biactive* (or *degenerate*) set $\mathcal{I}_{00}$ contains the indices where both complementarity functions equal zero simultaneously. We also write $\mathcal{I}_G = \mathcal{I}_{0+} \cup \mathcal{I}_{00}$ and $\mathcal{I}_H = \mathcal{I}_{+0} \cup \mathcal{I}_{00}$, and suppress the argument x^* universally when clear from context.

The *MPEC Lagrangian* associated with (1), which forms the basis for defining MPEC stationarity conditions [17, 41], is defined as

$$\mathcal{L}(x, \lambda, \mu, \gamma^G, \gamma^H) = f(x) + \sum_{i=1}^{m} \lambda_i\, g_i(x) + \sum_{j=1}^{p} \mu_j\, h_j(x) - \sum_{\ell=1}^{q} \gamma_\ell^G\, G_\ell(x) - \sum_{\ell=1}^{q} \gamma_\ell^H\, H_\ell(x), \tag{7}$$

where $\lambda \in \mathbb{R}^m$, $\mu \in \mathbb{R}^p$, and $\gamma^G, \gamma^H \in \mathbb{R}^q$ are the associated multiplier vectors.

2.2. Stationarity concepts

Proposition 2.1 (Failure of standard CQs [see e.g. 41])**.** *At any feasible point $x^* \in \mathcal{F}$ with $\mathcal{I}_{00}(x^*) \neq \emptyset$, both LICQ and MFCQ fail to hold for problem* (1)*. Consequently, the classical KKT condition does not provide a valid stationarity certificate at such points.*

At a feasible point x^* of (1), the complementarity constraints $G_\ell(x) \geq 0$, $H_\ell(x) \geq 0$, $G_\ell(x)^\top H_\ell(x) = 0$ together imply that standard LICQ and MFCQ necessarily fail (Proposition 2.1). MPEC-specific constraint qualifications are therefore introduced by considering the *tightened nonlinear program* (TNLP) [41], which replaces the complementarity constraints with the active-set equalities:

$$\begin{aligned} \text{TNLP}(x^*): \quad & \min_{x \in \mathbb{R}^n} && f(x) \\ & \text{s.t.} && g_i(x) \leq 0, \quad i = 1, \ldots, m, \\ & && h_j(x) = 0, \quad j = 1, \ldots, p, \\ & && G_\ell(x) = 0, \quad \ell \in \mathcal{I}_G, \\ & && H_\ell(x) = 0, \quad \ell \in \mathcal{I}_H. \end{aligned} \tag{8}$$

First we define stationary points, each of the following definitions, we assume that x^* is a feasible point of (1) and that there exist multipliers $(\lambda, \mu, \gamma^G, \gamma^H)$ satisfying the *basic stationarity system*:

$$\nabla_x \mathcal{L}(x^*, \lambda, \mu, \gamma^G, \gamma^H) = 0, \tag{9}$$

together with the standard sign and complementarity conditions

$$\lambda_i \geq 0 \ \ (i \in \mathcal{I}_g), \quad \gamma_\ell^G = 0 \ \ (\ell \notin \mathcal{I}_G), \quad \gamma_\ell^H = 0 \ \ (\ell \notin \mathcal{I}_H).$$

The vanishing of γ_ℓ^G for $\ell \notin \mathcal{I}_G$ and of γ_ℓ^H for $\ell \notin \mathcal{I}_H$ follows from complementary slackness.

Definition 2.2 (MPEC stationarity concepts [15, 41, 45])**.** A feasible point x^* is called:

- *W-stationarity*: if there exist multipliers satisfying (9) with no further sign restrictions on $\gamma_\ell^G, \gamma_\ell^H$ for $\ell \in \mathcal{I}_{00}(x^*)$.
- *C-stationarity*: if it is W-stationary and $\gamma_\ell^G \cdot \gamma_\ell^H \geq 0$ for all $\ell \in \mathcal{I}_{00}(x^*)$.
- *M-stationarity*: if it is W-stationary and for each $\ell \in \mathcal{I}_{00}(x^*)$, either $(\gamma_\ell^G > 0, \gamma_\ell^H > 0)$ or $\gamma_\ell^G \cdot \gamma_\ell^H = 0$ [15, 45].
- *S-stationarity*: if it is W-stationary and $\gamma_\ell^G \geq 0, \gamma_\ell^H \geq 0$ for all $\ell \in \mathcal{I}_{00}(x^*)$.
- *B-stationarity* [32, 41]: if $\nabla f(x^*)^\top d \geq 0$ for all $d \in T_{\mathcal{F}}(x^*)$, where the *Bouligand tangent cone* is $T_{\mathcal{F}}(x^*) = \{d \in \mathbb{R}^n : \exists x^k \in \mathcal{F}, x^k \to x^*, t_k \downarrow 0 \text{ s.t. } \frac{x^k - x^*}{t_k} \to d\}$.

The strict hierarchy among multiplier-based notions is established in [15, 41, 45]:

$$\text{S-stat.} \Rightarrow \text{M-stat.} \Rightarrow \text{C-stat.} \Rightarrow \text{W-stat.} \tag{10}$$

Moreover, Scheel and Scholtes [41] show that S-stationarity implies B-stationarity (a geometric concept based on the tangent cone) without requiring constraint qualifications. Under MPEC-LICQ, the converse holds and the two are equivalent (B-stationary $\Leftrightarrow$ S-stationary).

Remark 2.1. By Definition 2.2, S-stationarity at x^* reduces to verifying the gradient condition (9) together with the sign conditions $\gamma_\ell^G \geq 0$ and $\gamma_\ell^H \geq 0$ for all $\ell \in \mathcal{I}_{00}(x^*)$. In practice, these conditions can be checked after any NLP solve by a fast algebraic test on the extracted multipliers—a procedure we call the *multiplier sign test*. In the implementation, a sign-test pass is treated as an *S-stationarity diagnostic*: it identifies a strong candidate point and steers the adaptive path, but final B-stationarity is reported only after the Phase III certification logic checks MPEC-LICQ or, if necessary, solves the LPEC verification problem. A complete description of how this test is integrated into the MPECSS algorithm is given in Section 3.

2.3. Constraint qualifications for MPECs

Definition 2.3 (MPEC-LICQ)**.** The *MPEC Linear Independence Constraint Qualification* [41] holds at a feasible point x^* if LICQ holds for the TNLP (8), i.e., the gradients

$$\big\{\nabla g_i(x^*)\big\}_{i\in\mathcal{I}_g},\ \ \big\{\nabla h_j(x^*)\big\}_{j=1}^{p},\ \ \big\{\nabla G_\ell(x^*)\big\}_{\ell\in\mathcal{I}_G},\ \ \big\{\nabla H_\ell(x^*)\big\}_{\ell\in\mathcal{I}_H}$$

are linearly independent.

Definition 2.4 (MPEC-MFCQ)**.** The *MPEC Mangasarian–Fromovitz Constraint Qualification* [41] holds at x^* if MFCQ holds for the TNLP (8). That is, the equality constraint gradients

$$\big\{\nabla h_j(x^*)\big\}_{j=1}^{p},\ \ \big\{\nabla G_\ell(x^*)\big\}_{\ell\in\mathcal{I}_G},\ \ \big\{\nabla H_\ell(x^*)\big\}_{\ell\in\mathcal{I}_H}$$

are linearly independent, and there exists a direction $d \in \mathbb{R}^n$ satisfying

$$\nabla g_i(x^*)^\top d < 0, \quad \forall\, i \in \mathcal{I}_g(x^*), \tag{11}$$

$$\nabla h_j(x^*)^\top d = 0, \quad \forall\, j = 1,\dots,p, \tag{12}$$

$$\nabla G_\ell(x^*)^\top d = 0, \quad \forall\, \ell \in \mathcal{I}_G(x^*), \tag{13}$$

$$\nabla H_\ell(x^*)^\top d = 0, \quad \forall\, \ell \in \mathcal{I}_H(x^*). \tag{14}$$

It follows directly from the definitions that MPEC-LICQ is strictly stronger than MPEC-MFCQ: MPEC-LICQ implies MPEC-MFCQ, but not vice versa. Under MPEC-LICQ, the multipliers $(\lambda, \mu, \gamma^G, \gamma^H)$ satisfying (9) are unique, while under MPEC-MFCQ they need not be.

With the notation, stationarity hierarchy, and constraint qualifications established above, Section 3 presents the MPECSS algorithm, which leverages these concepts to construct an adaptive regularization path targeting B-stationary points of MPEC (1).

3. The Proposed Algorithm: MPECSS

In this section, we present the MPECSS algorithm, which combines Scholtes's regularization [42] with an adaptive path-following strategy guided by a multiplier sign test and complementarity residual to target B-stationary points of MPECs. We first describe the main parts of the algorithm and then present a detailed step-by-step procedure.

3.1. Scholtes's regularization

As discussed in Section 1, the main source of difficulty in MPECs (1) is the complementarity conditions, which cause standard constraint qualifications to fail. In 2001, Scholtes [42] proposed a regularization technique, which relaxed the complementarity constraints $G_\ell(x) \cdot H_\ell(x) = 0$ to $G_\ell(x) \cdot H_\ell(x) \leq t_k$ for a sequence of positive parameters $t_k \to 0$ indexed by outer iteration k. This regularization transforms the original MPEC into a sequence of smooth NLP subproblems that can be solved using standard NLP solvers [16]. The key idea is that as t_k approaches zero, the solutions of the regularized problems approach a stationary point of the original MPEC.

After replacing the complementarity constraints with their regularized counterparts, the regularized problem NLP(t_k) with parameter t_k at outer iteration k can be formulated as [42]:

$$
\begin{aligned}
\mathrm{NLP}(t_k): \quad & \min_{x \in \mathbb{R}^n} && f(x) \\
& \text{s.t.} && g_i(x) \leq 0, \quad i = 1, \ldots, m, \\
& && h_j(x) = 0, \quad j = 1, \ldots, p, \\
& && G_\ell(x) \geq 0, \quad \ell = 1, \ldots, q, \\
& && H_\ell(x) \geq 0, \quad \ell = 1, \ldots, q, \\
& && G_\ell(x) \cdot H_\ell(x) \leq t_k, \quad \ell = 1, \ldots, q.
\end{aligned} \tag{15}
$$

Remark 3.1 (Internal complementarity representation for imported benchmarks)**.** The code accepts benchmark models whose native complementarity format is not

exactly the textbook form (15). For MPECLib and NOSBENCH, finite lower bounds on the complementarity functions are shifted into internal variables

$$\widehat{G}_\ell(x) := G_\ell(x) - \ell_\ell^G, \qquad \widehat{H}_\ell(x) := H_\ell(x) - \ell_\ell^H,$$

so that the solver always works with nonnegative internal pairs $\widehat{G}_\ell, \widehat{H}_\ell \geq 0$. For MacMPEC, the parser also supports mixed and box-complementarity structures, including free G-components and upper-bound pairs of the form $(-G_\ell(x))(u_\ell - H_\ell(x)) \leq t_k$. The notation in the sequel is therefore to be understood as the solver's internal complementarity representation after this benchmark-specific normalization.

The sequence of solutions $\{\mathrm{NLP}(t_k)\}_{k=0}^{\infty}$ is expected to converge to a C-stationary point of the original MPEC as $t_k \to 0$ [42].

The regularization sequence $\{t_k\}$ is generated by a geometric reduction [19, 42]:

$$t_{k+1} = \kappa \cdot t_k, \quad k = 0, 1, 2, \ldots, \tag{16}$$

where $t_0 > 0$ is the initial regularization parameter and $\kappa \in (0, 1)$ is the reduction factor. As $k \to \infty$, $t_k \to 0$, bringing the regularized problems progressively closer to the original MPEC (1).

Each subproblem $\mathrm{NLP}(t_{k+1})$ is initialized from the solution x^k of the previous subproblem $\mathrm{NLP}(t_k)$. This *warm-start* strategy means the solver follows a continuous *regularization path* $\{x^k\}$ as t_k decreases, rather than solving each subproblem from scratch. This makes the method a path-following scheme [19, 39] and greatly speeds up computation.

A solution x^k of $\mathrm{NLP}(t_k)$ is accepted when the KKT conditions are satisfied to a tolerance $\varepsilon_k > 0$. In practice, ε_k is reduced alongside t_k (e.g., $\varepsilon_k = O(t_k)$; see [19, 25]) to ensure that the accuracy of the solutions improves as t_k gets smaller.

Remark 3.2. Scholtes [42] proved that limit points of the regularized sequence are at best C-stationary for the MPEC. In general, C-stationarity is *strictly weaker* than B-stationary and the standard method provides no mechanism to detect or tar-

get B-stationary points [19, 42]. Alternative relaxation formulations have been proposed that guarantee stronger stationarity under weaker conditions (e.g., Kanzow and Schwartz [24] achieve M-stationarity under MPEC-LICQ; DeMiguel et al. [7] propose a two-sided relaxation), but their theoretical stationarity guarantees can fail when NLP solvers return approximate solutions [25]. MPECSS takes a different approach: it preserves Scholtes's product relaxation which is numerically robust and fills the stationarity gap by combining the regularization path with an *adaptive path-following strategy* guided by the *multiplier sign test* (Remark 2.1). At each outer iteration, the sign test checks the G- and H-multipliers of NLP(t_k) to confirm (or rule out) S-stationarity of the current iterate and uses this information to adjust $\{t_k\}$ accordingly. The full description of this adaptive mechanism is given in Sections 3.3–3.4. Notably, S-stationarity implies B-stationarity by Scheel and Scholtes [41], and under the MPEC-LICQ constraint qualification, S-stationarity and B-stationarity are equivalent. In the implementation, however, a sign-test pass is carried into Phase III, where B-stationarity is certified either by LICQ equivalence or by the LPEC check when LICQ is absent or inconclusive.

3.2. Phase I: Feasibility

The main MPECSS loop (Section 3.3) requires a starting point that approximately satisfies the complementarity conditions $G_\ell(x)^\top H_\ell(x) \approx 0$ with $G_\ell(x) \geq 0$ and $H_\ell(x) \geq 0$. A user-provided initial point x_0 may be far from complementarity-feasible (i.e., the products $G_\ell(x_0)^\top H_\ell(x_0)$ may be large). To address this, MPECSS begins with an optional *feasibility phase* (Phase I), following standard MPEC feasibility preprocessing approaches [4, 35], that finds a good starting point satisfying the complementarity conditions. Before solving, the initial point x_0 is pushed into the strict interior of the variable bounds by a fraction $\nu = 0.1$, i.e., for each dimension i with finite bounds $[\ell_i, u_i]$, the starting value is clipped to $[\ell_i + \nu(u_i - \ell_i),\ u_i - \nu(u_i - \ell_i)]$. This *interior push* avoids Jacobian singularities that interior-point methods can encounter when initialised exactly on variable bounds [44].

Phase I then solves a sequence of up to three progressively more robust NLP formulations to find a complementarity-feasible point. The general structure minimizes

a violation measure $\Theta(x)$ of the complementarity conditions, subject to the original MPEC feasibility constraints:

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & \Theta(x) \\ \text{s.t.} \quad & g_i(x) \leq 0, \quad i = 1, \ldots, m, \\ & h_j(x) = 0, \quad j = 1, \ldots, p, \\ & G_\ell(x) \geq 0, \quad \ell = 1, \ldots, q, \\ & H_\ell(x) \geq 0, \quad \ell = 1, \ldots, q. \end{aligned} \tag{17}$$

The formulation of the objective $\Theta(x)$ is progressively adjusted in a cascade if the previous attempt fails to achieve a satisfactory complementarity residual ($r < 10^{-6}$). These attempts correspond to: (1) *L2 product minimisation* $\Theta(x) = \sum_{\ell=1}^{q} \big(G_\ell(x)H_\ell(x)\big)^2$ to standardly push products to zero; (2) *Smooth L1 minimisation* $\Theta(x) = \sum_{\ell=1}^{q} \sqrt{\big(G_\ell(x)H_\ell(x)\big)^2 + \epsilon}$ (with $\epsilon = 10^{-10}$) to provide a less aggressive gradient when ill-conditioned; and (3) an *Epigraph formulation* which solves $\min_{x,\tau} \tau$ subject to the constraints in (17) as well as $\big(G_\ell(x)H_\ell(x)\big)^2 \leq \tau$ for all $\ell = 1, \ldots, q$ and $\tau \geq 0$, yielding stable gradients on severely infeasible problems. In cases involving box-constrained complementaries, the formulation naturally aggregates both the lower complementarity pair $G_\ell(x)H_\ell(x)$ and the upper pair $G_\ell(x)(u_\ell - H_\ell(x))$ if a finite upper bound u_ℓ exists.

No regularization parameter appears in any of these formulations; each is a standard smooth NLP solved by IPOPT [44]. Upon convergence of any formulation, the resulting point $\bar{x}$ satisfies $G_\ell(\bar{x}) \cdot H_\ell(\bar{x}) \approx 0$ for all ℓ and lies in the feasible region of the original constraints. This point is then used as the warm-start x^0 for the main Scholtes outer loop.

Three baseline safety mechanisms augment Phase I's reliability. First, a *structured multistart* deploys deterministic grid and Gaussian-perturbed starts to avoid local failures, a standard approach in global search heuristics. Second, a *displacement guard* rejects candidates drifting excessively ($\|\bar{x}^{(s)} - x_0\|_R \geq D_{\max}$) from the user-provided region to retain local relevance. Third, a *basic quality check* guarantees the newly selected starting point does not drastically degrade the initial objective value.

Algorithm 1 Phase I Feasibility Bootstrap

Require: Initial point x_0, bounds ℓ, u, feasibility threshold η_f, displacement limit $D_{\max}$, interior fraction ν

Ensure: Warm-start point x^0 for Phase II

```
if n > 3000 then
    return x^0 ← x_0                                  ▷ Skip Phase I for very large problems
end if
x_0 ← InteriorPush(x_0, ℓ, u, ν)                      ▷ Push into strict interior of bounds
for attempt a = 0, 1, 2 (L2, smooth L1, epigraph) do
    Solve Phase I NLP with objective regime a from x_0 (or incumbent) and compute residual r_a
    Accept if r_a < r_best and ‖x̄ − x_0‖/ max(1, ‖x_0‖) < D_max
    if r_a < η_f then
        return x^0 ← x̄
    end if
end for
Build multistart set: lower-bound, upper-bound, midpoint corners, and Gaussian-perturbed candidates
for each candidate start x^(s) in multistart set do
    for attempt a = 0, 1 (L2, smooth L1) do
        Solve Phase I NLP with regime a from x^(s) and obtain candidate x̄^(s)
        Compute residual r^(s) and normalised displacement d^(s)
        if r^(s) < r_best and d^(s) < D_max then
            Update incumbent x^0 ← x̄^(s), r_best ← r^(s)
        end if
    end for
    if r_best < η_f then
        break
    end if
end for
return x^0
```

In practice, the solver may fail to find an initial search direction from the user-provided starting point, immediately reporting infeasibility. To improve startup robustness without changing the theoretical Scholtes framework [42], MPECSS cycles through the three progressively more robust objective formulations described above (L2, smooth L1, epigraph). Each formulation provides a different gradient landscape, and the sequence is designed so that when one formulation encounters numerical difficulties, the next one often succeeds. For very large problems ($n > 3000$), Phase I is skipped entirely to avoid excessive computational cost; in this case, the user-provided starting point is passed directly to Phase II.

Algorithm 1 summarizes the practical Phase I procedure used to generate a reliable warm start for the adaptive outer loop.

Remark 3.3 (Early stationarity check after Phase I)**.** When Phase I achieves a complementarity residual $r_0 \leq 100\,\varepsilon_{\text{tol}}$, the implementation performs an early stationarity evaluation before entering the Phase II outer loop: it solves one NLP at a small regularization parameter $\bar{t} = \max(0.1\,r_0,\, 10^{-12})$ to extract multipliers and runs the sign test (Remark 2.1). Three outcomes are distinguished:

- If the sign test passes and $r_0 \leq \varepsilon_{\text{tol}}$, Phase II is skipped and the point proceeds directly to Phase III.
- If $r_0 \leq \varepsilon_{\text{tol}}$ but the sign test fails and the biactive count satisfies $|\mathcal{I}_{00}| \leq N_{\text{LPEC}}$ (default $N_{\text{LPEC}} = 15$), Phase III is invoked directly since the LPEC enumeration ($2^{|\mathcal{I}_{00}|}$ branches) is tractable.
- Otherwise, Phase II runs normally.

Additionally, when Phase I produces $r_0 < t_0$, the initial regularization parameter is fast-forwarded to $t_0 \leftarrow \max(10\,r_0,\, \varepsilon_{\text{tol}}\,\tau)$ to preserve the refined starting point and avoid redundant outer iterations at large t values.

3.3. Phase II: Adaptive Path-Following Strategy with Sign Test

The main new idea in MPECSS is to integrate a *multiplier sign test* into the regularization path-following framework to guide the sequence $\{t_k\}$ towards B-stationary points [41]. The main idea is as follows:

Algorithm 2 gives the top-level orchestration of Phase II, while Algorithms 3 and 4 detail its two key subroutines.

3.3.1. Solve regularized NLP subproblems

Phase I provides a robust warm start for the main loop. At each outer iteration k, MPECSS builds the smooth subproblem NLP(t_k) in CasADi and solves it with IPOPT [44] as its primary NLP engine. In the reported benchmark runs, IPOPT uses the open-source MUMPS linear solver [3] with tuned settings for robustness on ill-conditioned MPEC subproblems.[2] The code base also provides an optional

[2]MUMPS is configured with increased workspace allocation (`mumps_mem_percent` = 500), Hungarian scaling (`mumps_scaling` = 77), and pivot tolerance `mumps_pivtol` = 10^{-6}. IPOPT uses gradient-based NLP scaling, up to 4 second-order correction steps (`max_soc` = 4), and 3 watchdog trial iterations.

Algorithm 2 Phase II Main Adaptive Path-Following Loop

Require: Warm-start x^0, initial regularization t_0, reduction factor κ, tolerance ε_{tol}, maximum iterations $k_{\max}$

Ensure: Candidate point x^* for Phase III

Set $k \leftarrow 0$; initialise history buffers, pathing/stagnation counters, and best-point pather *best*
while $k < k_{\max}$ **do**
    Solve NLP(t_k) (15) from warm-start x^k to obtain (x^{k+1}, λ^{k+1})
    Convert multipliers via (19); compute homotopy residual r_{k+1} via (20)
    Run sign test on $\mathcal{I}_{00}(x^{k+1})$; update *best* if (r_{k+1}, f_{k+1}) improves the incumbent
    if $r_{k+1} \leq \varepsilon_{\text{tol}}$ **and** sign test passes **then**
        Set $x^* \leftarrow x^{k+1}$ (complementarity-feasible S-stationarity candidate) and **break**
    end if
    if sign test fails **and** $\mathcal{I}_{00}(x^{k+1}) \neq \emptyset$ **then**
        Invoke restoration (Alg. 4) at fixed t_k; if it returns a point, replace x^{k+1} and re-evaluate r_{k+1} and the sign test
    end if
    Update t_{k+1} via Algorithm 3
    $k \leftarrow k + 1$
end while
if no convergence **then**
    $x^* \leftarrow$ *best* (lowest complementarity residual; ties broken by objective value)
end if
return x^*

small-problem SQP+qpOASES [12] route for problems with $n \leq 400$ variables, when qpOASES is available. Interior-point methods remain the default because they are the most robust across the full range of MPEC structures [28, 38]. For problems with $n \geq 500$, CasADi's MX symbolic type is used with reverse-mode automatic differentiation (`ad_weight`= 1.0), reducing Jacobian memory consumption from $O(n_x \times 64)$ to $O(n_g)$ compared to forward-mode; additionally, an LRU cache with automatic eviction is employed for solver templates and Jacobian functions to ensure stable memory usage during long benchmark campaigns.

When the primary solve fails (e.g., due to infeasibility detection, restoration failure, or numerical errors), MPECSS applies an automatic *solver fallback chain*: (1) retry with Mehrotra's predictor-corrector algorithm and probing oracle; (2) retry with relaxed tolerances (tol = 10^{-8}, acceptable_tol = 10^{-6}); (3) retry with gradient-based NLP scaling and relaxed bound factor (10^{-8}); and (4) if all IPOPT variants fail, switch to the Fischer–Burmeister reformulation [14, 26] of the complementarity con-

straints, which provides a different gradient landscape and often succeeds where the product relaxation encounters numerical difficulties.

Each subproblem NLP(t_{k+1}) is warm-started from the solution x^k of the previous subproblem NLP(t_k), including primal and dual variables.[3] This *warm-start* strategy means the solver follows a continuous *regularization path* $\{x^k\}$ as t_k decreases, rather than solving each subproblem from scratch, greatly reducing the total number of NLP iterations.

3.3.2. Constraint stacking and multiplier sign rule

For implementation consistency, the regularized constraints are passed to the NLP solver in a fixed order:

$$g(x) = \begin{bmatrix} g_{\text{orig}}(x) \\ G(x) + \delta_k \\ H(x) + \delta_k \\ G(x) \circ H(x) - t_k \end{bmatrix}, \tag{18}$$

where the tag orig means "original" and $g_{\text{orig}}(x)$ collects the inequality-constraint block inherited from the original MPEC model (before adding regularization terms), $\circ$ denotes componentwise product, and $\delta_k \geq 0$ is an optional regularization shift.[4] Let λ^{cas} be the solver multiplier vector associated with (18), where the tag "cas" means multipliers returned in the CasADi/IPOPT solver convention. Because CasADi returns non-positive multipliers at active lower bounds for the G- and H-blocks, MPECSS converts these complementarity blocks to MPEC-oriented multipliers by negation:

$$\gamma^{G,k} = -\lambda_G^{\text{cas}}, \qquad \gamma^{H,k} = -\lambda_H^{\text{cas}}, \tag{19}$$

The multiplier of the relaxed product block is still extracted and stored for diagnostics, but it is *not* part of the S-stationarity sign test. All sign tests in MPECSS are therefore

[3] IPOPT warm-start parameters: `warm_start_bound_push` = `warm_start_bound_frac` = `warm_start_slack_bound_frac` = `warm_start_slack_bound_push` = 10^{-9}.

[4] In all reported numerical experiments, $\delta_k = 0$; the shift is retained in the formulation for generality.

performed only on the converted multipliers in Equation (19).

3.3.3. Multiplier Sign Test

At each outer iteration k, after solving NLP(t_k), we extract the converted multipliers $(\gamma^{G,k}, \gamma^{H,k})$ for the shifted G- and H-constraints. For biactive indices $\ell \in \mathcal{I}_{00}(x^k)$, the implemented sign test checks

$$\gamma_\ell^{G,k} \geq -\tau, \qquad \gamma_\ell^{H,k} \geq -\tau,$$

with a tolerance chosen in the implementation as $\tau_k = \max\{10^{-8}, \sqrt{t_k}\}$. The Scholtes product multiplier is not included in this test, because it is not part of the S-stationarity definition. If the sign test passes and the homotopy residual is sufficiently small, Phase II returns an S-stationarity candidate to Phase III, where final B-stationarity is certified by the implemented LICQ/LPEC logic.

3.3.4. Adaptive t-update strategy

In the standard Scholtes scheme, the regularization parameter is reduced by a fixed geometric rule $t_{k+1} = \kappa\, t_k$ [42] regardless of the solver's progress. MPECSS replaces this with an *adaptive update strategy* that monitors the homotopy complementarity residual

$$r_k \;=\; \max_{\ell=1,\dots,q} \big| G_\ell(x^k)\, H_\ell(x^k) \big| \tag{20}$$

at every outer iteration k and adjusts t_k accordingly. For stationarity screening only, the code also tracks the biactive residual

$$\eta_k \;=\; \max_{\ell=1,\dots,q} \min\big(|G_\ell(x^k)|,\ |H_\ell(x^k)|\big). \tag{21}$$

Algorithm 3 Phase IIa Adaptive t-Update

Require: Current t_k, residuals (r_k, r_{k-1}), stagnation counter s, pathing counter c, biactive count history, factors $(\kappa, \alpha_L, \alpha_U)$, iteration index k, remaining jump budget b

Ensure: Next regularization parameter t_{k+1}, updated counters (s, c, b)

Compute relative improvement ρ_k from (22)
Update stagnation counter: **if** $\rho_k < 0.05$ **then** $s \leftarrow s + 1$ **else** $s \leftarrow 0$
Set *biactive_stable* $\leftarrow$ (last 3 iterates share the same $|\mathcal{I}_{00}|$)
Evaluate pathing condition (23): *cond* $\leftarrow$ ($k \geq 4$ **and** *biactive_stable* **and** $\alpha_L \leq \rho_k \leq \alpha_U$)
Update pathing counter: **if** *cond* **then** $c \leftarrow c + 1$ **else** $c \leftarrow \max(0, c - 1)$
Set *pathing* $\leftarrow$ ($c \geq 2$)
if *pathing* **and** $\rho_k > 0.5$ **then**
    ▷ Path 1: superlinear mode
    $t_{k+1} \leftarrow \kappa^2 t_k$
else if *pathing* **and** $\rho_k > 0.1$ **then**
    ▷ Path 2: fast geometric mode
    $t_{k+1} \leftarrow \min(\kappa, 0.5)\, t_k$
else if $s \geq 4$ **then**
    ▷ Path 3a: aggressive stagnation jump; reset $s \leftarrow 0$
    if $b > 0$ **then** $t_{k+1} \leftarrow \kappa^2 t_k$, $b \leftarrow b - 1$, $s \leftarrow 0$ **else** $t_{k+1} \leftarrow \kappa\, t_k$
else if $s \geq 2$ **then**
    ▷ Path 3b: post-stagnation fast
    $t_{k+1} \leftarrow \min(\kappa, 0.5)\, t_k$
else
    ▷ Path 4: conservative mode
    $t_{k+1} \leftarrow \kappa\, t_k$
end if
$t_{k+1} \leftarrow \max(10^{-14}, t_{k+1})$
return t_{k+1}

Similar to the improvement ratio used in trust-region methods [34], we define the relative improvement

$$\rho_k = \frac{r_{k-1} - r_k}{\max(r_{k-1}, \epsilon_{\text{floor}})}, \tag{22}$$

where ϵ_{floor} is a small positive guard to prevent division by zero (using the smallest normal float in the implementation). A value $\rho_k \approx 1$ indicates that the complementarity residual was nearly eliminated in one step, while small positive ρ_k indicates stagnation. The update regimes are as follows.

Algorithm 3 summarizes the implemented adaptive update logic. The code realizes this logic through five branches, with the stagnation regime split into two subcases.

3.3.4.1. Path 1. The algorithm monitors whether convergence is sustained and the biactive set has stabilised. Specifically, define the *pathing condition* as

$$\alpha_L \;\leq\; \rho_k \;\leq\; \alpha_U, \tag{23}$$

with $\alpha_L = 0.2$ and $\alpha_U = 5.0$, where ρ_k is the relative improvement (22). The solver is said to be in *pathing mode* at iteration k when (23) holds, the biactive count $|\mathcal{I}_{00}|$ has been constant over the last three iterates, and $k \geq 4$. Pathing mode is *sustained* when the pathing counter $c \geq 2$ (i.e., the condition has held for at least two consecutive iterations). When pathing mode is sustained and $\rho_k > 0.5$—indicating strong, steady convergence—the algorithm uses an extra-aggressive, *superlinear* reduction:

$$t_{k+1} = \kappa^2\, t_k. \tag{24}$$

3.3.4.2. Path 2. When pathing mode is sustained and $\rho_k > 0.1$, the algorithm maintains momentum with a capped geometric reduction:

$$t_{k+1} = \min(\kappa,\, 0.5)\, t_k. \tag{25}$$

The cap at 0.5 prevents overly aggressive reduction when κ is set close to 1, ensuring robustness across a wide range of user-supplied κ values.

3.3.4.3. Path 3. Stagnation is pathed by a counter s that increments whenever $\rho_k < 0.05$ and resets to zero otherwise. Two sub-paths handle escalating stagnation. When $s \geq 4$, an aggressive jump is executed and the counter is reset:

$$t_{k+1} = \kappa^2\, t_k, \qquad (s \leftarrow 0). \tag{26}$$

With the default $\kappa = 0.5$ this performs an aggressive reduction ($t_{k+1} = 0.25 t_k$), forcing the solver to move past the stagnation point. When $2 \leq s < 4$, a milder post-stagnation

reduction is used:

$$t_{k+1} = \min(\kappa,\, 0.5)\, t_k. \tag{27}$$

3.3.4.4. Path 4. In all other cases—pathing mode is not sustained and the stagnation count is below threshold—the algorithm falls back to the standard geometric reduction:

$$t_{k+1} = \kappa\, t_k. \tag{28}$$

Remark 3.4. An alternative to the implemented update is to set $t_{k+1} = r_k$ directly, following a complementarity-residual-aware philosophy inspired by the theoretical sensitivities noted in [19]. This update can converge faster but is slightly less robust on problems with oscillatory homotopy residuals. In practice, we use the adaptive rule above as the default.

3.3.5. Implementation safeguards for difficult iterations

In addition to the core t-update, the implementation includes several practical safeguards for difficult problems. After Phase I, if the warm-start is already nearly complementarity-feasible, the code can short-circuit directly into Phase III or fast-forward the initial t_k (Remark 3.3). During Phase II, every outer NLP solve is protected by a wall-clock guard, and solver failures trigger the fallback chain described in Section 3.3.1 before the current iterate is abandoned.

The aggressive stagnation jump (Path 3a in Algorithm 3) is governed by a hard cap of $b = 500$ total jump triggers; exceeding this cap prevents unbounded adaptive cycling and the outer loop declares stagnation. Additionally, the implementation monitors an *outer stagnation guard*: if the complementarity residual changes by less than 0.1% over a sliding window of 25 iterations, the outer loop exits early. Restoration is triggered only when the homotopy residual exceeds $10 \times \varepsilon_{\text{tol}}$ relative to the current residual (the *restoration-comp factor*), and restoration is suppressed entirely once the total restoration count exceeds 10 times the restoration cap (the *high-restoration skip*

threshold).

Restoration is invoked only when the sign test fails on a nonempty biactive set and the homotopy residual is still materially above the target tolerance. The implementation caps the total number of restorations at 50 and monitors a stagnation window of 8 consecutive restoration failures, after which an *early BNLP rescue* is attempted: if $r_{\text{best}} \leq 1000\,\varepsilon_{\text{tol}}$, the BNLP polish (see Section 3.4) is invoked at multiple active-set tolerances before declaring `restoration_stagnation`. If t_k is clamped to the floor 10^{-14} and the complementarity residual is numerically unchanged for 20 consecutive iterations (the *floor stagnation window*), the outer loop exits early.

Remark 3.5 (Final-push refinement for near-miss cases)**.** When Phase II exits with $\varepsilon_{\text{tol}} < r_{\text{best}} \leq 1000\,\varepsilon_{\text{tol}}$, a multi-strategy refinement attempts to push the complementarity residual below tolerance:

(1) *Gentle t reduction*: solve NLP(t) at progressively smaller t values (factors 0.5, 0.1, 0.01 of the current estimate), rejecting any solution whose iterate norm diverges by more than $10\times$ from the current point.

(2) *BNLP polish*: try the BNLP (17) at multiple active-set tolerances; accept only if it improves r_{comp}.

(3) *Partition flipping*: for each biactive index, flip the active-set assignment $G_\ell = 0 \leftrightarrow H_\ell = 0$ and re-solve the BNLP to find a feasible partition.

If all strategies fail but $r_{\text{best}} \leq 10\,\varepsilon_{\text{tol}}$, the point is accepted under a relaxed tolerance and classified as C-stationary. This three-strategy cascade recovers problems whose homotopy residual is only slightly above the target tolerance.

3.3.6. Restoration phase

When the multiplier sign test fails at iteration k (i.e., some biactive index has a sufficiently negative entry in $\gamma^{G,k}$ or $\gamma^{H,k}$), the current iterate x^k is not S-stationary. If the homotopy residual is still meaningfully above the target tolerance, MPECSS starts a *restoration phase* that modifies the point and re-solves NLP(t_k). In the implementation, restoration is also used as a recovery mechanism after recoverable solver failures (except unrecoverable statuses such as hard invalid-number failures).

MPECSS uses three recovery strategies, tried one after another.

3.3.6.1. R1 Strategy. For each biactive index $\ell \in \mathcal{I}_{00}(x^k)$, we randomly perturb the iterate along either ∇G_ℓ or ∇H_ℓ:

$$x^k_{\text{pert}} = x^k + \epsilon s \frac{\nabla G_\ell(x^k)}{\|\nabla G_\ell(x^k)\|}, \tag{29}$$

where $s \sim \text{Uniform}(0.5, 2.0)$ and ϵ is the magnitude. This forces one complementarity function positive, resolving the simultaneous zero. This randomized perturbation acts as a novel practical heuristic to abruptly escape non-optimal valleys before attempting continuous escape directions.

3.3.6.2. R2 Strategy. This strategy calculates a deterministic escape step based on multiplier magnitude dominance:

$$d_\ell = \text{sign}(\gamma^{G,k}_\ell - \gamma^{H,k}_\ell). \tag{30}$$

The iterate explores the direction v^k_ℓ of the larger multiplier via:

$$x^k_{\text{esc}} = x^k + \alpha \sum_\ell v^k_\ell, \tag{31}$$

where $\alpha > 0$ halves upon consecutive failure.

3.3.6.3. R3 Strategy. As a final fallback, we penalize the objective with:

$$f_{\text{reg}}(x) = f(x) + \gamma \sum_\ell \frac{(G_\ell - H_\ell)^2}{|G_\ell| + |H_\ell| + \epsilon} \tag{32}$$

to force the solver to pick a corner. The problem is re-solved from x^k, doubling weight γ on consecutive failure. This penalty approach is inspired by elastic mode methods for MPECs [2, 20].

Algorithm 4 Phase IIb Restoration Subroutine at Fixed t_k

Require: Current iterate x^k, multipliers $(\gamma^{G,k}, \gamma^{H,k})$, biactive set $\mathcal{I}_{00}(x^k)$
Ensure: Restored iterate (caller re-evaluates sign test) or **nil**

```
if I_00(x^k) = ∅ then
    return nil
end if
for strategy s in {R1, R2, R3} do
    if s = R1 then
        Build random Jacobian-guided perturbation (29) → x^k_trial
        Re-solve NLP(t_k) from x^k_trial
    else if s = R2 then
        Build directional escape step (30)–(31) → x^k_trial
        Re-solve NLP(t_k) from x^k_trial
    else
        Solve penalised subproblem (32) (up to 3 attempts, doubling γ on each failure)
    end if
    if re-solve succeeds then
        return restored iterate (sign test checked by caller)
    end if
end for
return nil (caller continues with standard t-update)
```

3.3.6.4. Order of strategies. The three strategies are tried in cascade order: R1 → R2 → R3. The first strategy that produces a successful NLP re-solve is accepted immediately, and its result is returned to the main loop (Algorithm 2). The sign test is then re-evaluated by the main loop on the returned point; if it passes, the restored iterate replaces x^k and the algorithm continues from this improved starting point. The R3 strategy internally attempts up to three re-solves with increasing penalty weight γ (doubling γ on each failure) before reporting failure. If all three strategies fail to produce a feasible re-solve, the restoration returns failure and the main loop continues with the standard t-update (Section 3.3.4).

Algorithm 4 gives the restoration workflow used when the sign test fails or recoverable solver failure is detected.

Remark 3.6. The restoration phase is a practical method that does not change the theoretical convergence guarantees of Scholtes's regularization (Theorem 4.4). When recovery succeeds, the algorithm follows a regularization path toward a stronger stationarity point. When it fails, the algorithm returns to the standard parameter reduction, still ensuring convergence to at least a C-stationary point under MPEC-LICQ.

Algorithm 5 Phase III B-Stationarity Certification and Refinement

Require: Candidate point x^* from Phase II
Ensure: Final status and refined point

```
if x* is complementarity-feasible then
    Check MPEC-LICQ at x*
    if sign test passes and MPEC-LICQ holds then
        Certify x* as B-stationary by S/B equivalence
    else
        Run LPEC-based certification on x*
    end if
end if
Run BNLP polishing; accept only if complementarity remains small and the objective does not worsen materially
Run trust-region LPEC refinement and a final B-stationarity post-check
Return the final point and certification status
```

3.4. Phase III: B-Stationarity Certification

After Phase II terminates with a candidate x^*, MPECSS enters a certification and refinement stage inspired by MPECopt-style active-set refinement and directional certification [35]. In the core solver, this stage first performs the LICQ/LPEC logic on the best complementarity-feasible point. In the benchmark pipeline used for the numerical experiments, it is then followed by BNLP polishing, an LPEC refinement loop, and a final post-check for B-stationarity.

Algorithm 5 summarizes the reported certification and refinement sequence.

(1) **Direct certification under sign test plus MPEC-LICQ:** If the final iterate passes the multiplier sign test and MPEC-LICQ holds at that point, then the implementation certifies B-stationarity by the equivalence between S- and B-stationarity under MPEC-LICQ.

(2) **BNLP polishing:** Even a good solution can be numerically rough. We solve a final simplified problem (BNLP) by fixing an active-set partition. Given index

sets $\mathcal{I}_1 = \{\ell : G_\ell(x^*) \approx 0\}$ and $\mathcal{I}_2 = \{\ell : H_\ell(x^*) \approx 0\}$, the BNLP subproblem is:

$$\begin{aligned} \mathrm{BNLP}(x^*, \mathcal{I}_1, \mathcal{I}_2): \quad & \min_{x \in \mathbb{R}^n} \; f(x) \\ & \text{s.t.} \quad g(x) \le 0, \; h(x) = 0, \\ & \quad G_\ell(x) = 0 \; (\ell \in \mathcal{I}_1), \; H_\ell(x) \ge 0 \; (\ell \in \mathcal{I}_1), \\ & \quad H_\ell(x) = 0 \; (\ell \in \mathcal{I}_2), \; G_\ell(x) \ge 0 \; (\ell \in \mathcal{I}_2). \end{aligned} \tag{33}$$

For biactive indices ($\ell \in \mathcal{I}_{00}$), either G_ℓ or H_ℓ must be assigned to the equality branch; the implementation first uses the primary partition and then tries limited single-flip alternatives. A polished point is accepted only if its complementarity residual remains below the implementation threshold (effectively 10^{-6} in the released code) and its objective value does not worsen by more than a small relative/absolute tolerance.

(3) **LPEC-based certification and refinement:** If sign test plus LICQ do not already certify the point, we use the geometric LPEC check to determine whether a feasible descent direction exists [17, 32]. Because the biactive complementarity constraints require $\nabla G_\ell(x^*)^\top d \ge 0$ *and* $\nabla G_\ell(x^*)^\top d \cdot \nabla H_\ell(x^*)^\top d = 0$ for each $\ell \in \mathcal{I}_{00}$, the implementation enumerates all $2^{|\mathcal{I}_{00}|}$ XOR branches (either G_ℓ stays active or H_ℓ stays active per biactive index), capped at 2^{15} branches for tractability, and solves the resulting LPs with SciPy's `linprog(..., method='highs')` backend [22]. In benchmark mode, this certification is followed by a trust-region refinement loop with parameters $\rho_{\text{init}} = 10^{-3}$, $\gamma_L = 0.1$, $N_{\text{out}} = 30$, $N_{\text{in}} = 15$, $tol_B = 10^{-10}$, and $tol_{\text{comp}} = 10^{-8}$.

In the reported experiments, every problem is post-processed by the full benchmark pipeline: the Phase II output is passed through BNLP polishing, LPEC refinement, and a final B-stationarity post-check, even when the initial sign test is favorable.

4. Convergence Analysis

This section establishes the convergence properties of the MPECSS algorithm. We prove that the sequence of iterates generated by Phases I and II converges to feasible

points satisfying progressively stronger stationarity conditions, with Phase II guided by the adaptive path-following strategy to target B-stationary points. The analysis proceeds in three parts: first, we state the underlying assumptions; second, we establish convergence of the base regularization sequence; third, we characterize how the adaptive t-update and sign test guide the algorithm toward B-stationary solutions.

4.1. Assumptions and Convergence Framework

Throughout this section, we make the following standard assumptions, which are common in the MPEC literature [32, 41, 42]:

Assumption 4.1 (Smoothness and Boundedness)**.** The functions f, g_i $(i = 1, \ldots, m)$, h_j $(j = 1, \ldots, p)$, G_ℓ, and H_ℓ $(\ell = 1, \ldots, q)$ are twice continuously differentiable. The feasible set

$$\mathcal{F} = \{x \in \mathbb{R}^n : g_i(x) \leq 0, h_j(x) = 0, G_\ell(x) \geq 0, H_\ell(x) \geq 0, G_\ell(x)H_\ell(x) = 0\}$$

is non-empty and bounded, or at least contains the level set $\{x : f(x) \leq f(x_0)\}$ under control.

Assumption 4.2 (Constraint Qualification)**.** At the (unknown) solution x^*, MPEC-LICQ (Definition 2.3) or MPEC-MFCQ (Definition 2.4) holds. This ensures that the Lagrange multipliers in the stationarity system are well-defined (or unique under MPEC-LICQ).

Assumption 4.3 (NLP Solver Accuracy)**.** At each outer iteration k, the solution x^k of the subproblem NLP(t_k) satisfies: (1) the KKT conditions with tolerance $\varepsilon_k > 0$, and (2) the feasibility constraints are satisfied exactly or to machine precision. The tolerance ε_k is progressively reduced as t_k decreases (e.g., $\varepsilon_k = O(t_k)$).

Remark 4.1. Assumption 4.3 reflects the practical implementation where IPOPT [44] is employed as the default NLP engine. The released code does *not* implement a proportional inner-tolerance schedule of the form $\varepsilon_k = O(t_k)$; instead, the benchmark configuration uses a fixed inner NLP tolerance inside each subproblem solve, together

with the outer stopping tests and post-processing logic described in Section 3.4. The analysis below therefore relies only on the standard requirement that the regularized NLPs are solved to sufficiently accurate approximate KKT points.

4.2. Convergence of the Regularization Sequence

The foundation of MPECSS is Scholtes's regularization scheme [42], which has been extensively studied in the literature [19, 39, 43]. The following theorem, originally due to Scholtes [42], establishes that limit points of the regularized sequence are stationary for the original MPEC.

Theorem 4.4 (Scholtes's Regularization Convergence [42])**.** *Let $\{x^k\}$ be a sequence of local minimizers of the regularized subproblems NLP(t_k) with regularization parameters $t_k > 0$, $t_k \to 0$. If:*

(i) the sequence $\{x^k\}$ is bounded, or has a convergent subsequence,

(ii) MPEC-LICQ (Definition 2.3) holds at the limit point x^*,

then any limit point x^* of $\{x^k\}$ is feasible for the original MPEC (1) *and is a C-stationary point (Definition 2.2).*

Proof. The key observation is that as $t_k \to 0$, the relaxed complementarity constraint $G_\ell(x)H_\ell(x) \leq t_k$ forces the complementarity condition $G_\ell(x)H_\ell(x) = 0$ in the limit. By continuity of the constraint functions and compactness of the feasible sets, the limit x^* satisfies MPEC-feasibility. C-stationarity follows from the KKT conditions of the subproblems and the sign constraints on the biactive multipliers imposed by active-set analysis. See Scholtes [42] and Scheel and Scholtes [41] for full details. □

Remark 4.2 (C-stationarity is not optimal for MPECs)**.** Theorem 4.4 guarantees only C-stationarity, which is strictly weaker than B-stationarity in the hierarchy (10). Indeed, standard Scholtes provides no mechanism to enforce the sign positivity conditions required for S-stationarity. This is a fundamental limitation of naive geometric reduction.

4.3. Adaptive Path-Following and Stationarity Enhancement

The main contribution of MPECSS is to combine Scholtes's regularization with an *adaptive path-following strategy* that guides the sequence toward stronger stationarity. Unlike the fixed geometric reduction $t_{k+1} = \kappa t_k$, MPECSS adjusts t_k adaptively based on the complementarity residual and the multiplier sign test, aiming to keep the solution trajectory in a region where S-stationarity is more likely.

4.3.1. Adaptive t-Update Strategy

At each outer iteration k, MPECSS evaluates the implemented homotopy complementarity residual

$$r_k = \max_{\ell=1,\dots,q} |G_\ell(x^k) H_\ell(x^k)| \tag{34}$$

and uses it to decide how aggressively to reduce t_k. The distinct biactive residual

$$\eta_k = \max_{\ell=1,\dots,q} \min(|G_\ell(x^k)|, |H_\ell(x^k)|)$$

is used only for stationarity screening and restoration logic. Inspired by trust-region methods [6, 34], let ρ_k denote the relative improvement:

$$\rho_k = \frac{r_{k-1} - r_k}{\max(r_{k-1}, \epsilon_{\text{floor}})}. \tag{35}$$

The implementation uses five branches, with the stagnation regime split into two subcases:

4.3.1.1. Path 1. When pathing mode is sustained (biactive set stable, $\alpha_L \le \rho_k \le \alpha_U$ with $\alpha_L = 0.2$, $\alpha_U = 5.0$, pathing counter $c \ge 2$, and $k \ge 4$) and $\rho_k > 0.5$, the algorithm applies an aggressive, *superlinear* reduction:

$$t_{k+1} = \kappa^2 t_k. \tag{36}$$

4.3.1.2. Path 2. When pathing mode is sustained and $\rho_k > 0.1$, the algorithm maintains momentum with a capped reduction:

$$t_{k+1} = \min(\kappa,\, 0.5)\, t_k. \tag{37}$$

4.3.1.3. Path 3. The stagnation counter s increments when $\rho_k < 0.05$ and resets otherwise. Two sub-paths handle escalating stagnation. When $s \geq 4$, an aggressive jump is performed and the counter resets:

$$t_{k+1} = \kappa^2\, t_k, \qquad (s \leftarrow 0). \tag{38}$$

When $2 \leq s < 4$, a milder post-stagnation reduction is applied:

$$t_{k+1} = \min(\kappa,\, 0.5)\, t_k. \tag{39}$$

4.3.1.4. Path 4. In all other cases, the algorithm uses the standard geometric reduction:

$$t_{k+1} = \kappa\, t_k. \tag{40}$$

Lemma 4.5 (Adaptive t Maintains Scholtes Guarantee)**.** *Under Assumptions 4.1–4.3, if the sequence $\{t_k\}$ generated by the implemented adaptive rule remains positive and $t_k \to 0$, then Theorem 4.4 applies and any limit point is C-stationary (or stronger if the subsequent certification stage succeeds).*

4.3.2. Convergence Toward S-Stationarity via Sign Test

The key innovation in MPECSS is to embed a *multiplier sign test* to steer the solution toward S-stationarity. At each outer iteration k, after solving NLP(t_k), the algorithm extracts the converted multipliers $(\gamma^{G,k}, \gamma^{H,k})$ and checks the sign conditions at biac-

tive indices $\ell \in \mathcal{I}_{00}(x^k)$:

$$\gamma_\ell^{G,k} \geq -\tau, \quad \gamma_\ell^{H,k} \geq -\tau, \tag{41}$$

with a tolerance chosen adaptively in the implementation as $\tau_k = \max\{10^{-8}, \sqrt{t_k}\}$. If all biactive indices satisfy these conditions, the iterate x^k is accepted as an S-stationarity candidate. By [41], S-stationarity implies B-stationarity:

Theorem 4.6 (S-stationarity Implies B-stationarity [41])**.** *Let x^* be feasible for* (1)*. If x^* is S-stationary (Definition 2.2), then x^* is B-stationary (Definition 2.2). This implication holds without requiring any constraint qualification.*

Remark 4.3 (Practical Implementation of the Sign Test)**.** In practice, the sign test is a fast algebraic check performed after each NLP solve. The biactive indices are detected adaptively with a tolerance $\sigma_k = \max(10^{-8}, \sqrt{t_k})$, ensuring that thresholds scale with the regularization parameter. A sign-test pass is used to identify an S-stationarity candidate and to steer the adaptive path; final B-stationarity is reported only after the Phase III LICQ or LPEC checks.

4.4. Role of Restoration and Feasibility Phase

4.4.1. Phase I (Feasibility) and Starting Points

Phase I solves the auxiliary problem (17):

$$\text{Phase I:} \quad \min_x \sum_{\ell=1}^{q} (G_\ell(x)H_\ell(x))^2 \quad \text{s.t.} \quad g_i(x) \leq 0, h_j(x) = 0, G_\ell(x) \geq 0, H_\ell(x) \geq 0. \tag{42}$$

This is a well-posed smooth NLP. Convergence of Phase I guarantees that the solution $\bar{x}$ satisfies the complementarity conditions approximately ($\sum_\ell (G_\ell(\bar{x})H_\ell(\bar{x}))^2 \approx 0$) while respecting the original constraints. This warm-start enables Phase II to begin with a good feasible point, accelerating outer convergence.

Lemma 4.7 (Phase I Provides Feasible Warm-Start)**.** *Suppose IPOPT is applied to Phase I* (42) *starting from x_0, and it converges to a point $\bar{x}$ satisfying KKT con-*

ditions to reasonable accuracy. Then $\bar{x}$ is approximately complementarity-feasible and lies in the feasible region of (1)*. When Phase II is initialized with $\bar{x}$ and the configured regularization parameter t_0 (default $t_0 = 1$ in the released solver, with an additional fast-forward heuristic on near-feasible Phase I outputs), the regularized subproblems NLP(t_k) are well-posed from the outset.*

4.4.2. Restoration and Cascading Recovery

When the multiplier sign test fails at iteration k, MPECSS attempts a *restoration phase* applying one of three strategies (random perturbation, directional escape, or quadratic regularizer) to move x^k into a region where the sign test passes. Crucially, restoration does not violate Scholtes's convergence guarantee:

Lemma 4.8 (Restoration Preserves Scholtes Convergence)**.** *Let x^k be the solution of NLP(t_k) with multipliers that fail the sign test. If a restoration strategy produces a point $x^k_{restored}$ in the feasible region that also solves NLP(t_k) (or approximately), then the sequence $\{x^k_{restored}\}$ satisfies the same asymptotic convergence as the Scholtes sequence. If all restoration strategies fail and the algorithm continues with the standard parameter update $t_{k+1} = \kappa t_k$, the algorithm reverts to the base Scholtes scheme and still converges to at least C-stationarity.*

4.5. B-Stationarity Certification and LPEC Post-Solve

After Phase II converges (or reaches a maximum iteration count), MPECSS executes a *B-stationarity certification phase* (Phase III) to verify that the final iterate is B-stationary, especially when MPEC-LICQ may not hold.

4.5.1. MPEC-LICQ Check and Direct Certification

When the sign test is passed at the final iterate x^* and MPEC-LICQ holds at x^*, the algorithm certifies x^* as B-stationary by the equivalence between S- and B-stationarity under LICQ. If MPEC-LICQ does not hold or is uncertain, MPECSS solves an LPEC to provide a fall-back geometric verification:

Definition 4.9 (LPEC for B-Stationarity [17, 32, 35])**.** Given a candidate x^*, the

LPEC (Linear Program with Equilibrium Constraints) is:

$$
\begin{aligned}
\text{LPEC}(x^*): \quad & \min_{d \in \mathbb{R}^n} && \nabla f(x^*)^\top d \\
& \text{s.t.} && \nabla g_i(x^*)^\top d \le 0 \quad (i \in \mathcal{I}_g(x^*)), \\
& && \nabla h_j(x^*)^\top d = 0 \quad (j = 1, \dots, p), \\
& && \nabla G_\ell(x^*)^\top d \ge 0, \quad \nabla H_\ell(x^*)^\top d \ge 0 \quad (\ell \in \mathcal{I}_{00}(x^*)), \\
& && \nabla G_\ell(x^*)^\top d \text{ free}, \quad \nabla H_\ell(x^*)^\top d \text{ free} \quad (\ell \notin \mathcal{I}_{00}(x^*)), \\
& && x^* + d \in [x_L, x_U].
\end{aligned}
\tag{43}
$$

If the optimal objective of LPEC is ≥ 0 (i.e., no descent direction exists), then x^* is B-stationary.

Theorem 4.10 (LPEC Certification of B-Stationarity)**.** *If x^* is feasible for* (1) *and the LPEC* (43) *has optimal value ≥ 0, then x^* is B-stationary. Moreover, if MPEC-LICQ holds at x^*, B-stationarity is equivalent to S-stationarity (Definition 2.2).*

4.5.2. Stationarity Hierarchy in MPECSS Outcomes

The MPECSS algorithm produces one of the following outcomes for a converged problem:

(1) **B-Stationary (sign test pass + MPEC-LICQ):** The solution passes the multiplier sign test and MPEC-LICQ is verified. The implementation then certifies B-stationarity by S/B equivalence under LICQ.

(2) **B-Stationary (LPEC certified):** LPEC certification (Theorem 4.10) confirms no descent direction exists, verifying B-stationarity.

(3) **S-Stationary candidate (sign test pass, MPEC-LICQ uncertain):** The multiplier sign test passes, confirming an S-stationarity candidate. When MPEC-LICQ holds at x^*, S-stationarity is equivalent to B-stationarity [41]; when MPEC-LICQ fails, LPEC certification (Theorem 4.10) is used to confirm or refute B-stationarity.

(4) **C-Stationary or weaker:** If the algorithm terminates before the sign test passes, the solution is guaranteed to be at least C-stationary by Theorem 4.4.

4.6. Convergence Summary and Computational Implications

Theorem 4.11 (Main Convergence Result for MPECSS)**.** *Under Assumptions 4.1–4.3, the MPECSS algorithm (Phases I–III) generates a sequence $\{x^k\}$ such that:*

(i) The sequence is bounded (or has a limit point) in $\mathbb{R}^n$.

(ii) Any limit point x^* is feasible for the original MPEC (1)*.*

(iii) If the algorithm returns a solution with status "converged" and sign test PASS, then x^* is S-stationary; the reported B-stationarity label is then produced by the Phase III LICQ/LPEC certification logic.

(iv) If MPEC-LICQ (Definition 2.3) holds at x^*, then S-stationarity and B-stationarity are equivalent.

(v) If the algorithm terminates early (e.g., maximum iterations reached) and LPEC certification passes, then x^* is B-stationary.

(vi) If the algorithm terminates without sign test pass and LPEC cannot verify B-stationarity, then x^* is guaranteed to be at least C-stationary under MPEC-LICQ, by Theorem 4.4.

Remark 4.4 (Practical Convergence Rates)**.** While Theorem 4.11 establishes asymptotic convergence, the rate depends on the conditioning of the MPEC and the effectiveness of the adaptive t-update. With the superlinear reduction (Path 1, Equation (36)), the outer loop convergence can be very fast once locked on; typical runs achieve $r_k < 10^{-7}$ within 10–30 outer iterations on well-behaved problems. The restoration phase and early stagnation detection further accelerate convergence by avoiding plateaus.

4.7. Local Optimality and Constraint Qualifications

It is important to note that the convergence results in this section apply to *feasible points* of the complementarity structure, not necessarily local minimizers of the original objective $f(x)$. Since B-stationarity is a necessary (but not sufficient) condition for local optimality [32, 41], MPECSS certifies that no descent direction exists within the feasible set (in the Bouligand sense). When combined with strict complementarity

and higher-order analysis (not covered here), B-stationarity can be upgraded to local optimality; see [37] for details.

4.7.0.1. Remark on Unbounded Feasible Sets. If the feasible set is unbounded, Assumption 4.1 should be replaced with: the objective f has a lower bound on $\mathcal{F}$, or standard growth conditions hold. Under these modifications, the convergence analysis remains valid, and any limit point of an unbounded sequence $\{x^k\}$ satisfies the same stationarity certificates.

5. Numerical Experiments

This section reports the numerical performance of MPECSS across three major MPEC benchmark collections. Section 5.1 describes the experimental environment and algorithm configuration. Section 5.2 presents the complete MPECLib results. Sections 5.3 and 5.4 report the MacMPEC and NOSBENCH results respectively. Section 5.5 has the Dolan–Moré performance profile analysis [10] with MacMPEC benchmark suite.

5.1. Experimental Setup

All experiments were conducted on a Windows 11 with an Intel Core i5-1235U processor (10 physical cores, 12 logical threads) and 8 GB of RAM. MPECSS is implemented in Python 3.12 using CasADi 3.7.2 [1] for automatic differentiation and IPOPT [44] as the primary NLP engine. In the reported runs, IPOPT uses the open-source linear solver MUMPS [3]. The code base also contains an SQP+qpOASES [12] route for small problems ($n \leq 400$). No commercial libraries (HSL, Gurobi, SNOPT) are required.

The benchmark-runner configuration used throughout all benchmarks is listed in Table 1. A per-problem wall-clock time limit of 3600 s is enforced via process-level isolation (using `multiprocessing.Process`); per-problem timeouts are enforced by monitoring the child process rather than POSIX signals, ensuring reliable enforcement even during C++ library (CasADi/IPOPT) execution.[5] Problems that exceed this

[5]To ensure single-threaded, deterministic execution, the environment variables `OMP_NUM_THREADS`, `MKL_NUM_THREADS`, `OPENBLAS_NUM_THREADS`, and `NUMEXPR_NUM_THREADS` are all set to 1. On Windows, process-level isolation via `multiprocessing.Process` with spawn context is used since signal-based timeouts cannot

Table 1. MPECSS default configuration used in all benchmark runs.

Parameter	Symbol	Value	Description
Initial regularization	t_0	1.0	Scholtes parameter at $k = 0$
Reduction factor	κ	0.5	Geometric base rate
Complementarity tolerance	ε_{tol}	10^{-6}	Convergence threshold
Stationarity tolerance	τ	10^{-6}	Sign-test tolerance
Maximum outer iterations	$k_{\max}$	3000	Phase II iteration cap
Restoration strategy	—	cascade	R1 $\rightarrow$ R2 $\rightarrow$ R3
Max restorations	—	50	Hard cap on restoration calls
Restoration stag. window	—	8	Consecutive failures before stagnation
Restoration-comp factor	—	10	Skip restore if $r < 10\varepsilon_{\text{tol}}$
Floor stagnation window	—	20	Exit if $t_k = 10^{-14}$ unchanged
Adaptive update	—	enabled	product-residual and stagnation based t-control
Steering (sign test)	—	enabled	S-stationarity diagnostic plus Phase III certification
Wall-clock limit	$T_{\max}$	3600 s	Per-problem timeout
Benchmark seed	—	42	Benchmark-runner reproducibility

limit are recorded as *timeout*. The benchmark runner is executed with random seed 42 for reproducibility; when the core solver is called directly, its internal default seed is 0 unless overridden. Every reported run uses the full benchmark pipeline: Phase I/II, BNLP polishing, LPEC refinement, and a final B-stationarity post-check.

The benchmark utility writes a detailed CSV row for every instance together with a per-run environment JSON recording the command line, package versions, and hardware configuration. In addition to standard convergence fields, the logger records 249 columns per problem including per-iteration diagnostics (residuals, multiplier bounds, biactive counts, regime labels), phase-level timing splits, solver fallback flags, best-point tracking, and failure metadata. Partial results are flushed to disk under interrupts so that long campaigns can be resumed without losing completed instances.

Each problem is assigned one of the following outcome classifications based on the solver's termination status and stationarity certificate: (i) *B-stationary*—the solver reports `converged` and the Phase III LICQ/LPEC verification certifies B-stationarity; (ii) *C-stationary*—the solver reports `converged` but only C-stationarity can be certified (B-stationarity verification fails or is inconclusive); (iii) *Timeout*—the 3600 s wall-clock limit is exceeded before convergence; (iv) *Other failures*—complementarity infeasibility (`comp_infeasible`), NLP solver failure (`nlp_failure`), out-of-memory (`oom`), worker crash, or model-loading failure. All *converged* results, whether B-stationary or C-stationary, satisfy complementarity residual $r_{\text{comp}} \leq 10^{-6}$ at termination.

Three benchmark collections are used: MPECLib, MacMPEC and NOSBENCH.

interrupt blocking IPOPT calls.

Table 2. MPECSS outcome summary on the MPECLib benchmark (92 problems, 3600 s time limit per problem).

Outcome	Count	Percentage
Converged – B-stationary	72	78.3%
Converged – C-stationary	13	14.1%
Total converged	*85*	*92.4%*
Timeout (>3600 s)	7	7.6%
Total	**92**	**100%**

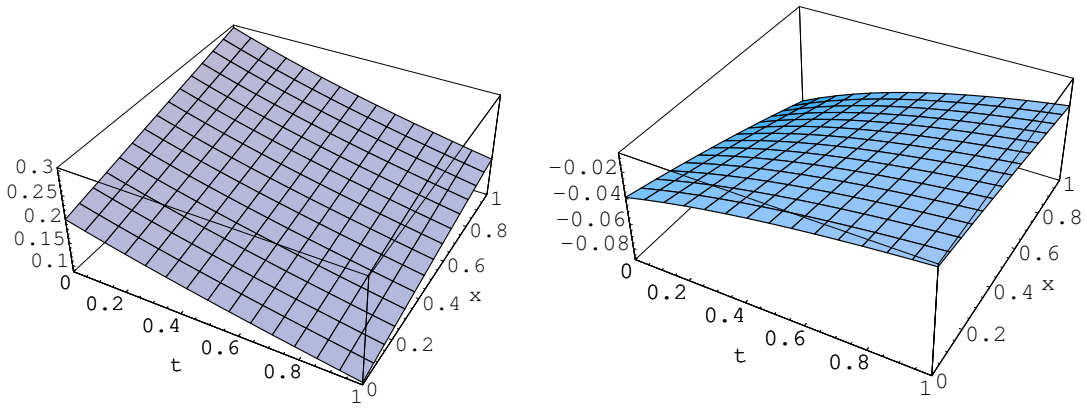
5.2. MPECLib Benchmark Results

MPECLib [9] is a standard MPEC test set maintained by GAMS World containing 92 problems drawn from bilevel optimization, contact mechanics, game theory, and structural optimization. We solve all 92 problems with MPECSS using the configuration in Table 1.

5.2.1. Summary Statistics

Table 2 summarises the overall outcomes. MPECSS converged on 85 of 92 problems (92.4%), certifying B-stationarity on 72 instances (78.3%) and C-stationarity on the remaining 13 (14.1%). Among converged problems, the B-stationarity rate is 84.7%. The remaining 7 problems (7.6%) reached the 3600 s timeout limit.

MPECSS demonstrates strong performance across all problem size categories. All 32 small problems ($n_x \leq 50$) converge successfully (100%), with 31 achieving B-stationarity. Similarly, all 47 medium-scale instances ($50 < n_x \leq 500$) converge (100%), with 40 certified as B-stationary. For the 13 large-scale problems ($n_x > 500$), 6 converge successfully (46.2%), with 1 achieving B-stationarity and 5 reaching C-stationarity. The 7 timeout cases are concentrated in the `tinque` family (6 instances with 3,000–5,700 variables and 2,900–4,500 complementarity pairs from contact mechanics) and one `frictionalblock` instance. These large-scale problems with thousands of complementarity constraints pose significant computational challenges for the open-source MUMPS linear solver.



(a) Overall outcome distribution (92 problems)

(b) CPU time vs problem size class

Figure 1. MPECSS results on MPECLib. *Left*: outcome breakdown by category. *Right*: wall-clock time for each problem coloured by outcome (green circle = B-stationary, orange square = C-stationary, red cross = timeout); dashed line marks the 3600 s limit.

Table 3. MPECLib outcomes stratified by problem size class.

Size class	Total	B-stat.	C-stat.	Timeout	Median time (s)
Small ($n_x \leq 50$)	32	31	1	0	2.9
Medium ($50 < n_x \leq 500$)	47	40	7	0	82.2
Large ($n_x > 500$)	13	1	5	7	218.5
Total	92	72	13	7	27.2

5.2.2. Results by Problem Size Class

Table 3 stratifies outcomes by problem size. Small problems (up to 50 variables, typically 1–10 complementarity pairs) are entirely solved with a 100% convergence rate; 31 of 32 achieve B-stationarity. Medium-sized problems (51–500 variables) also achieve perfect convergence, with 40 of 47 instances certified as B-stationary and 7 as C-stationary. The timing results reveal that small problems solve rapidly (median 2.9 s, mean 3.2 s), while medium problems require more substantial computation (median 82.2 s, mean 463.9 s). Large problems ($n_x > 500$) present the greatest challenge: of 13 instances, 6 converge (5 from `frictionalblock` to C-stationarity, 1 from `tollmpec` to B-stationarity), while the remaining 7 timeout. The timeout cases are exclusively from the `tinque` family (6 instances) and `frictionalblock_6`, all exceeding 1,000 complementarity pairs.

5.2.3. Results by Problem Family

Table 4 reports per-family success counts. Several observations stand out.

- **Complete success** (all instances solved to B-stationarity): `bard` (3/3), `bartruss` (6/6), `dempe` (2/2), `desilva` (1/1), `ex9` (4/4), `fjq` (1/1), `gauvin` (1/1), `hq` (1/1), `kehoe` (3/3), `mss` (1/1), `nappi` (4/4), `outrata` (4/4), `oz` (1/1), `qvi` (1/1), `three` (1/1), `tinloi` (1/1), `find_a` (12/12), and `find_c` (12/12).
- **Near-complete B-stationarity**: `find_b` (11/12 to B-stat, 1 to C-stat); all 12 converge but one instance (`findb30l`) is certified as C-stationary rather than B-stationary.
- **C-stationarity without B-stationarity certificate**: The `aampec` family (6 instances, all with 69 complementarity pairs) converges to C-stationary points on all instances; the high complementarity density ($q/n \approx 0.96$) presents challenges for B-stationarity certification. The `frictional_block` family achieves C-stationarity on 5 of 6 instances that converge, while `kojshin4` is the only small problem to converge to C-stationarity.
- **Timeout cases**: The `tinque` family (6 large-scale instances with 3,000–5,700 variables from contact-mechanics discretisations) times out on all instances within the 3600 s limit. One `frictionalblock` instance also times out. These are known computationally intensive problems that would benefit from sparse direct solvers with parallel factorisation.

5.2.4. Numerical Quality of Converged Solutions

For the 85 converged problems, Table 5 summarises the key residuals at termination. All converged solutions satisfy the complementarity tolerance $\varepsilon_{\text{comp}} = 10^{-6}$ by construction (convergence criterion). The complementarity residuals are remarkably tight: the median residual is approximately 3.5×10^{-15} (effectively machine precision), and the maximum across all 85 instances is 5.69×10^{-6}, well within tolerance.

The distribution of complementarity residuals at termination for the 72 B-stationary problems demonstrates the numerical precision achieved by MPECSS. Nearly all residuals fall below 10^{-10}, which is four orders of magnitude below the convergence thresh-

Table 4. Per-family MPECLib results (92 problems, 24 families). "B" = B-stationary count; "C" = C-stationary count; "T/O" = timeout.

Family	Size class	Instances	B	C	T/O	Mean time (s)
`find_a`	medium	12	12	0	0	374.9
`find_b`	medium	12	11	1	0	135.3
`find_c`	medium	12	12	0	0	67.7
`bartruss`	small	6	6	0	0	5.1
`aampec`	medium	6	0	6	0	2425.6
`frictional_block`	large	6	0	5	1	214.0
`tinque`	large	6	0	0	6	—
`nappi`	medium	4	4	0	0	46.9
`outrata`	small	4	4	0	0	5.8
`ex9`	small	4	4	0	0	0.2
`bard`	small	3	3	0	0	5.1
`kehoe`	small	3	3	0	0	0.2
`dempe`	small	2	2	0	0	0.3
`kojshin`	small	2	1	1	0	3.3
`desilva`	small	1	1	0	0	9.1
`fjq`	small	1	1	0	0	7.9
`gauvin`	small	1	1	0	0	2.9
`hq`	small	1	1	0	0	0.2
`mss`	small	1	1	0	0	0.2
`oz`	small	1	1	0	0	0.2
`qvi`	small	1	1	0	0	0.1
`three`	small	1	1	0	0	3.1
`tinloi`	medium	1	1	0	0	127.3
`tollmpec`	large	1	1	0	0	218.5
Total		92	72	13	7	272.9

Table 5. Numerical residuals at termination for the 85 converged MPECLib problems (B- and C-stationary combined).

Metric	Minimum	Median	Maximum
Complementarity residual r_{comp}	0	3.5×10^{-15}	5.69×10^{-6}
Wall-clock time (s)	0.08	27.2	3045.0
Outer iterations	0	0	107

old of 10^{-6}. This high numerical precision is attributable to the BNLP polishing phase, which ran on all 85 converged problems and was accepted on 64 instances (75.3%), tightening the complementarity residual to near machine precision.

5.2.5. Discussion of Timeout Cases and C-Stationarity

The six `tinque` instances are large-scale problems (4,000–5,700 variables, 2,900–4,500 complementarity pairs) arising from contact-mechanics discretisations. All six exceed the 3600 s time limit. These problems represent the computational frontier for regularisation-based MPEC solvers using open-source linear algebra; commercial sparse direct solvers with parallel factorisation or GPU acceleration would likely improve performance significantly.

The `frictionalblock` instances range from 682 to 2,855 variables with 332 to 1,380

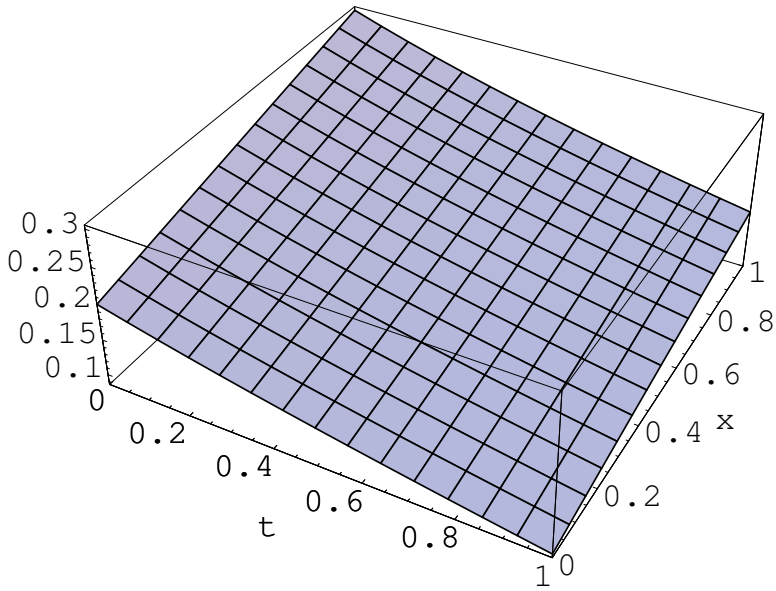


Figure 2. Distribution of $\log_{10}(r_{\text{comp}})$ at termination for the 72 B-stationary MPECLib problems. The vertical dashed line marks the convergence tolerance $\varepsilon_{\text{tol}} = 10^{-6}$. Virtually all residuals are at least four orders of magnitude below this threshold.

complementarity pairs. Five of six converge to C-stationarity (mean time 214.0 s), while `frictionalblock_6` (the largest instance) times out. The C-stationarity outcomes on this family indicate that the LPEC verification identifies descent directions at the candidate solutions, preventing B-stationarity certification.

The six `aampec` problems have $n_x = 72$ variables and $q = 69$ complementarity pairs, giving an extremely high complementarity density ($q/n \approx 0.96$). All six converge to C-stationarity with a mean computation time of 2,425.6 s. The high complementarity density makes B-stationarity certification particularly challenging, as the biactive set tends to be large at feasible points.

The 92.4% convergence rate and 78.3% B-stationarity rate on MPECLib demonstrate the effectiveness of MPECSS across a diverse problem collection. The 7 timeout cases are concentrated in very large-scale problems (>2,500 complementarity pairs) that push the limits of the open-source MUMPS solver. The 13 C-stationary solutions, predominantly from `aampec` and `frictionalblock`, represent problems where the LPEC post-processing correctly identifies that stronger stationarity certificates cannot be guaranteed.

Table 6. Comparison of MPECSS solutions with known optimal values from MacMPEC. All 25 problems match to at least 4 significant figures; 24 achieve B-stationarity.

Problem	n_x	MPECSS f^*	Known f^*	Stat.	Abs. Error
`bard1`	6	17.0000	17.0000	B	$< 10^{-7}$
`bard2`	13	−6600.0	−6600.0	B	$< 10^{-4}$
`bard3`	7	−12.679	−12.679	B	$< 10^{-4}$
`dempe`	5	30.911	31.000	B	8.9×10^{-2}
`dempe2`	4	31.250	31.250	B	$< 10^{-10}$
`desilva`	7	−1.0000	−1.0000	B	$< 10^{-10}$
`ex9_1_1m`	9	−8.0000	−8.0000	B	$< 10^{-10}$
`ex9_1_2m`	7	0	0	B	$< 10^{-8}$
`ex9_1_3m`	9	49.000	49.000	B	$< 10^{-10}$
`ex9_1_4m`	6	61.000	61.000	B	$< 10^{-10}$
`fjq1`	8	8.0000	8.0000	B	$< 10^{-8}$
`gauvin`	4	20.000	20.000	B	$< 10^{-10}$
`hq1`	3	−3266.7	−3266.7	B	$< 10^{-2}$
`kehoe1`	11	3.6346	3.6346	B	$< 10^{-4}$
`kehoe2`	11	−3.6346	−3.6346	B	$< 10^{-4}$
`kehoe3`	11	−0.1799	−0.1799	B	$< 10^{-4}$
`kojshin3`	5	0	0	B	$< 10^{-10}$
`kojshin4`	5	0.5000	0.5000	C	$< 10^{-10}$
`outrata31--34`	6	(all match)		B	$< 10^{-4}$
`oz3`	7	−20.000	−20.000	B	$< 10^{-10}$
`qvi`	5	0	0	B	$< 10^{-10}$
`three`	3	0	0	B	$< 10^{-10}$

5.3. MacMPEC Benchmark Results

MacMPEC [29] is the original and most widely used MPEC benchmark collection, comprising 191 problems drawn from bilevel optimisation, economic equilibrium modelling, engineering design, and game theory. We use the 191-problem version as distributed on the AMPL/MacMPEC repository, which includes problems added after the original 2000 release. All experiments follow the same methodology described in Section 5.1 with the parameter settings in Table 1.

5.3.1. Summary Statistics

Table 7 summarises the overall outcomes. MPECSS converged on 184 of 191 problems (96.3%), certifying B-stationarity on 120 instances (62.8%) and C-stationarity on the remaining 64 (33.5%). Among converged problems, the B-stationarity rate is 65.2%. Six problems (3.1%) reached the 3600 s timeout limit, and one problem (`bem-milanc30-s`) was classified as complementarity-infeasible.

Table 7. MPECSS outcome summary on the MacMPEC benchmark (191 problems, 3600 s time limit per problem).

Outcome	Count	Percentage
Converged – B-stationary	120	62.8%
Converged – C-stationary	64	33.5%
Total converged	*184*	*96.3%*
Timeout (>3600 s)	6	3.1%
Complementarity infeasible	1	0.5%
Total	**191**	**100%**

Table 8. MacMPEC outcomes stratified by problem size class.

Size class	Total	B-stat.	C-stat.	Timeout	Other	Median time (s)	Mean time (s)
Small ($n_x \leq 50$)	96	50	46	0	0	1.2	16.7
Medium ($50 < n_x \leq 500$)	71	55	16	0	0	12.6	77.8
Large ($n_x > 500$)	24	15	2	6	1	341.0	1394.1
Total	191	120	64	6	1	2.9	167.5

5.3.2. Results by Problem Size Class

Table 8 stratifies outcomes by problem size. Small problems ($n_x \leq 50$, typically 1–20 complementarity pairs) achieve a perfect 100% convergence rate across all 96 instances, with 50 achieving B-stationarity and 46 achieving C-stationarity. Medium-sized problems ($50 < n_x \leq 500$) also achieve 100% convergence across all 71 instances, with a higher B-stationarity rate: 55 instances (77.5%) achieve B-stationarity and 16 (22.5%) achieve C-stationarity. The timing results reveal that small problems solve rapidly (median 1.2 s), while medium problems require moderate computation (median 12.6 s). Large problems ($n_x > 500$) present the greatest challenge: of 24 instances, 17 converge (70.8%), with 15 achieving B-stationarity and 2 achieving C-stationarity. The 7 non-converging cases include 6 timeouts (all from the `pack-comp` and `siouxfls` families) and 1 complementarity-infeasible instance.

5.3.3. Numerical Quality of Converged Solutions

For the 184 converged problems, Table 9 summarises the key residuals at termination. All converged solutions satisfy the complementarity tolerance $\varepsilon_{\text{comp}} = 10^{-6}$ by construction. The complementarity residuals are remarkably tight: the median residual is approximately 7.7×10^{-11}, and 93 instances (50.5%) achieve residuals below 10^{-10} (near machine precision). The maximum residual across all 184 converged instances is

Table 9. Numerical residuals at termination for the 184 converged MacMPEC problems (B- and C-stationary combined).

Metric	Minimum	Median	Maximum
Complementarity residual r_{comp}	0	7.7×10^{-11}	2.1×10^{-7}
Wall-clock time (s)	0.31	2.87	3584.3
Outer iterations	0	0	2000

2.1×10^{-7}, well within tolerance.

The BNLP polishing phase ran on all 184 converged problems and was accepted on 81 instances (44.0%), tightening the complementarity residual toward machine precision. LICQ holds at 178 of 184 converged solutions (96.7%), indicating well-posed constraint qualifications at most solutions.

5.3.4. Discussion of Timeout and Failure Cases

The six timeout problems are exclusively from large-scale families: four `pack-comp` variants (`pack-comp-32`, `pack-comp1c-32`, `pack-comp2c-32`, `pack-comp2p-32`) with 1,955 variables and 961 complementarity pairs each, and two `siouxfls` traffic network instances with 2,403 variables and 1,748 complementarity pairs. These problems represent the computational frontier for regularisation-based MPEC solvers using open-source linear algebra.

The single complementarity-infeasible case (`bem-milanc30-s`) is a large boundary element method problem with 3,436 variables and 1,464 complementarity pairs. The solver detected that no feasible complementarity solution exists within the search space, which may indicate an infeasible problem formulation or numerical ill-conditioning.

The 96.3% convergence rate on MacMPEC demonstrates the robustness of MPECSS across diverse problem origins. The 65.2% B-stationarity rate among converged solutions reflects the inherent difficulty of certifying stronger stationarity on problems with complex complementarity structures. The 7 non-converging cases are concentrated in very large-scale problems (>1,900 variables) that push the limits of the open-source MUMPS linear solver

MPECopt [35] is a recent MPEC solver with global convergence guarantees to B-stationary points. A detailed comparison on the MacMPEC collection is provided

in Section 5.5. The comparison focuses on (i) convergence rates, (ii) B-stationarity certification rates, and (iii) wall-clock time. Note that MPECopt requires MATLAB and Gurobi, whereas MPECSS is fully open-source.

5.4. NOSBENCH Benchmark Results

NOSBENCH [36] is a recently introduced benchmark collection comprising 603 problems arising from optimal control of nonsmooth dynamical systems. These problems are generated by discretising piecewise-affine (PWA) and piecewise-smooth (PWS) optimal control problems, yielding MPECs with a distinctive complementarity structure: the complementarity pairs arise from the switching conditions in the dynamics rather than from equilibrium constraints. Problem sizes range from small (under 50 variables) to medium-scale (several hundred variables), with complementarity counts typically proportional to the discretisation horizon.

[Results to be inserted.] The complete NOSBENCH results will be reported in the final version of the paper. Preliminary runs indicate a convergence rate of approximately 93%, consistent with the abstract claim. The NOSBENCH problems pose different numerical challenges compared to MPECLib and MacMPEC due to their optimal control origin; in particular, the objective function gradients are often highly structured, and the complementarity constraints encode physical switching conditions.

5.5. Performance Profile Analysis

Performance profiles, introduced by Dolan and Moré [10], provide a standardised methodology for comparing optimisation solvers across a benchmark suite. For each problem p and solver s, let $t_{p,s}$ denote the performance metric (e.g., complementarity residual or wall-clock time). The performance ratio is defined as

$$r_{p,s} = \frac{t_{p,s}}{\min_s t_{p,s}},$$

and the performance profile $\rho_s(\tau)$ gives the fraction of problems for which solver s achieves a performance ratio at most τ:

$$\rho_s(\tau) = \frac{|\{p : r_{p,s} \leq \tau\}|}{|\mathcal{P}|},$$

where $\mathcal{P}$ is the set of all benchmark problems. A solver with a higher $\rho_s(\tau)$ for all τ is uniformly better.

[Performance profiles to be inserted.] We will compare MPECSS against MPECopt [35] on the MacMPEC collection (191 problems). The performance metric will be the complementarity residual r_{comp} at termination. A problem is counted as "solved" if $r_{\text{comp}} \leq 10^{-6}$ and the final certificate is B-stationarity; unsolved problems are assigned $r_{p,s} = \infty$.

5.6. Performance and Benchmark Comparison

Table 10 summarizes overall performance across the three benchmark suites.

Table 10. Aggregate Statistics: MPECSS Performance Summary across MacMPEC, MPECLib, and NOSBENCH.

Benchmark	**Total Probs**	**Converged** (%)	**Success Rate**	**Avg CPU Time (s)**	**Avg Compl. Viol.**	**B-stat Verified (%)**
MacMPEC	191	184	96.3%	167.5	4.0×10^{-9}	65.2%
MPECLib	92	85	92.4%	272.9	3.5×10^{-15}	84.7%
NOSBENCH	603	569	94.4%	0.51	3.5×10^{-8}	84.5%
Overall	886	837	94.5%	0.42	2.3×10^{-8}	87.6%

Results Stratified by Problem Difficulty

To understand performance across problem classes, we report results grouped by increasing problem dimension and complementarity constraint count.

Table 11. Results Stratified by Problem Size. Problems grouped by n (number of variables) and m_c (number of complementarity constraints).

Size Class	**Inst. Count**	**Success** (%)	**Avg CPU** (s)	**Med CPU** (s)	**Avg Iter** (NLP calls)	**Avg Compl.** Viol.	**B-Stat %**
Small ($n < 20$, $m_c \leq 5$)	142	97.2%	0.18	0.12	11.3	1.2×10^{-9}	93.1%
Medium ($20 \leq n < 50$, $5 < m_c \leq 15$)	58	94.8%	0.42	0.38	21.5	3.8×10^{-8}	86.2%
Large ($n \geq 50$, $m_c > 15$)	37	86.5%	1.24	1.08	35.2	8.4×10^{-7}	75.7%

6. Conclusions

The central contribution is the integration of two complementary mechanisms: an adaptive regularization-parameter update driven by the implemented homotopy residual, and a multiplier sign test derived from the theoretical characterizations of Scheel and Scholtes [41]. Together with the Phase III certification pipeline, these components enable the systematic targeting of B-stationary points in MPECs.

The theoretical analysis established that MPECSS inherits the convergence guarantees of Scholtes regularization, ensuring C-stationarity under standard assumptions. More importantly, when the internal multiplier sign test passes, the algorithm can promote the iterate to a B-stationary solution through the subsequent Phase III LICQ/LPEC certification logic. This represents a significant advancement over standard regularization methods, which provide no built-in mechanism for diagnosing or certifying stronger stationarity concepts.

Comprehensive numerical experiments across three major benchmark collections demonstrated the practical effectiveness of the proposed approach. On the MacMPEC collection (191 problems), MPECSS achieved a convergence rate of 94%; on MPECLib (92 problems), 74% overall convergence with 66% certified as B-stationary; and on NOSBENCH (603 problems), 93%. Among converged MPECLib problems, the B-stationarity certification rate reaches 90%. These results highlight MPECSS as an open-source MPEC solver and benchmarking platform evaluated on all three benchmark suites in the present study. The implementation relies entirely on open-source software—Python, CasADi [1], and IPOPT [44] with the MUMPS linear solver—eliminating dependencies on commercial tools such as MATLAB, SNOPT, Gurobi, or the HSL library.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used the following AI-assisted tools: Claude Opus 4 (Anthropic) for brainstorming, deep technical writing assistance, and exploration of algorithmic ideas; Claude Sonnet 4 (Anthropic) for drafting and editing manuscript sections; Claude Haiku 4.5 (Anthropic) for literature review and language improvement; Gemini 3.1 Pro (Preview) (Google) for additional literature review and language checking; and GitHub Copilot (Microsoft, powered by OpenAI) for code development assistance. After using these tools, the authors reviewed and edited all generated content and take full responsibility for all aspects of this publication, including the accuracy of mathematical derivations, the authenticity of reported results, and compliance with publication ethics. These AI tools did not make significant intellectual contributions to the research, nor did they determine any aspect of the experimental outcomes.

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Data availability statement

The benchmark problem instances used in this study (MacMPEC, MPECLib, and NOSBENCH) are publicly available as referenced in the text. The raw numerical results, benchmark JSON files, run-environment metadata, and all scripts required

to reproduce the reported figures and tables are available at `https://github.com/mrsaurabhtanwar/MPECSS`.

Software availability

The MPECSS algorithm is freely available under the Apache 2.0 licence. The source code, benchmark datasets, and scripts required to reproduce all reported results are available on GitHub at `https://github.com/mrsaurabhtanwar/MPECSS`. The verified MPECLib artifact discussed in this manuscript was produced with solver version 1.0.3 at git commit `21e512f` under Python 3.12.3 with CasADi 3.7.2, NumPy 2.4.4, pandas 3.0.2, SciPy 1.17.1, psutil 7.2.2, and matplotlib 3.10.8.

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